

Stochastic Quantization of the Chern–Simons Theory

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We discuss stochastic quantization of $d=3$ dimensional non-Abelian Chern–Simons theory. We demonstrate that the introduction of an appropriate regulator in the Langevin equation yields a well-defined equilibrium limit, thus leading to the correct propagator. We also analyze the connection between $d=3$ Chern–Simons and $d=4$ topological Yang–Mills theories, showing the equivalence between the corresponding regularized partition functions. Finally, we discuss the introduction of a non-trivial kernel as an alternative regularization. © 1992 Academic Press, Inc.

1. INTRODUCTION

The construction of quantum field theories which are independent of the space-time metric, topological quantum field theories (TQFTs), has shown to be a fruitful area of research, with applications both in mathematics and in physics [1–8] (see [4] for a complete list of references). In TQFTs there are no local excitations and then the only observables are topological invariants. In this way, Witten has shown how to obtain Donaldson invariants in four-dimensional manifolds [1], Jones polynomials of knot theory [2], etc. From the point of view of physics, TQFTs have been investigated in connection with two- and three-dimensional gravity models, two-dimensional conformal field theories, etc.

Quantization of topological field theories can be envisaged using different approaches. For example, the path-integral method leads to explicit expressions for the partition function and other topological invariants [1–6]. Alternatively, one can quantize in the functional Schrödinger representation [3], use canonical quantization for systems with constraints [2, 7] or stochastic quantization (SQ) [8]. These two last approaches have revealed, as a bonus, interesting connections between models in different number of space-time dimensions [9–11].

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Moreover, the SQ approach [12] to TQFTs has clarified the construction of supersymmetric models by exploiting the connection between Langevin equations and Nicolai maps which trivialize the respective models. This connection, introduced by Parisi and Sourlas [13] and Cecotti and Girardello [14] in low dimensions ($d < 4$) was extended to $d \geq 4$ when quantization of TQFTs was undertaken using the Batalin–Vilkovisky approach [5, 6].

Many aspects related to SQ of TQFTs remain to be fully understood. It is the purpose of the present work to address to some of them. In particular, we carefully discuss the problem of convergence towards equilibrium of the stochastic process associated to non-Abelian Chern–Simons theory. To this end, we introduce an appropriate regulator in the associated Langevin equation showing that the resulting perturbative stochastic diagrams lead to the set of ordinary Feynman diagrams for non-Abelian Chern–Simons theory. We also show that the connection between $d=3$ Chern–Simons model and $d=4$ topological Yang–Mills (TYM) theory, which was originally established [9–11] without taking care of the convergence problem, still holds when a regulating scheme is taken into account.

To more precisely understand the problems to be discussed, let us briefly review in this Introduction the basic features arising in the quantization of TQFTs.

First, since the action $S[\phi]$ for fields ϕ in TQFTs does not involve at all the metric of the space-time manifold, several problems regarding quantization are posed. Within the path-integral approach, the metric could enter through BRST terms needed to fix its large (*topological*) symmetry. One should then be sure that the topological character of the theory is not spoiled by quantization. In order to show that the partition function is indeed metric independent, one has to specify an invariant measure (being the naïve measure metric dependent). This can be done by taking as integration variables not the original fields but appropriate tensorial densities [15–17]. In this way, one can easily show that the topological character of the model is preserved. The same care regarding integration variables has to be taken in defining the partition function for TQFTs within the SQ approach. This problem has been analyzed in detail in Ref. [11].

There are, however, new problems in the SQ approach that are associated with TQFTs peculiarities. In particular, being the action $S[\phi]$ metric independent, it remains unchanged when passing from Minkowski to Euclidean space. This poses, of course, a problem since SQ was originally formulated for real Euclidean actions. The problem of how to handle complex actions has nevertheless been carefully analyzed [18, 19] and it has recently received much attention [20, 21].

In a particularly attractive TQFT, and here we come to a main point in this paper, there is another source of problems conspiring against convergence towards equilibrium within the stochastic approach. Indeed, consider three-dimensional Chern–Simons (CS) theory. Since the corresponding action is linear in derivatives one should expect non-convergence of the associated stochastic process towards equilibrium. In order to solve this problem, refined treatments for the Langevin equation of a CS theory have been proposed. They basically rest on the introduction of non-trivial kernels [21, 22] (this compares to the treatment of fermionic

models) both for Abelian and non-Abelian cases or Maxwell terms [21], for the Abelian case, as regulators. In both cases it has been proven that conventional propagators can be obtained very simply thus reproducing the standard results for Chern–Simons theories.

As stated above, for *non-Abelian* CS theory, convergence of the stochastic process has been only discussed by introducing an appropriate kernel [21]. This approach poses several difficulties when studying the connection between $d=3$ CS theory and TYM theory. We shall then follow an alternative regularization approach by extending the method employed in [21] (based on the introduction of a regulating Maxwell term for the Abelian theory) to the non-Abelian case. We shall analyze the convergence of the associated stochastic process towards equilibrium and we shall briefly comment on how the connection between $d=3$ CS theory and $d=4$ TYM theory is established in the presence of the regulator.

The paper is organized as follows: in Section 2 we introduce a $\text{tr } F_{ij}^2$ regulating term in the Langevin equation and show, up to second order in perturbation theory, that stochastic diagrams reproduce the standard Feynman rules for CS theory. We then reanalyze the connection between $d=3$ CS and $d=4$ TYM theories and give a survey on topological features of the resulting four-dimensional effective theory. We also discuss in this section the alternative regularization scheme based on the introduction of a non-trivial kernel. Finally, Section 3 is devoted to the conclusions.

2. REGULATING THE STOCHASTIC PROCESS

As we stated in the Introduction, SQ of $d=3$ Chern–Simons theory faces problems of convergence towards equilibrium when the naïve Langevin equation is used. These problems are rooted in two main features of topological field theories. First, being the corresponding action metric-independent, it remains unchanged when passing from Minkowski to Euclidean space, leading to a purely imaginary Euclidean action. Second, being the CS theory linear in its derivatives, it does not ensure a finite equilibrium limit.

Wu and Zhu [21] have discussed how to handle both problems. On the one hand, one can employ the so-called complex Langevin equation approach [18]: no inconsistency then arises due to the presence of the i -factor in front of the Euclidean CS action. Concerning the second point, these authors introduce as a regulator a F_{ij}^2 term when studying the Abelian theory. At least in this last case, one ends up with a stochastic process tending to equilibrium and leading to standard results for propagators once the regulator is switched off.

In this section we shall discuss this problem of convergence for the *non-Abelian* Chern–Simons theory within the SQ scheme. To this end we introduce a $\text{tr } F_{ij}^2$ term and then analyze the convergence of the stochastic process (at the end of this section we discuss the alternative method of introducing a kernel in the Langevin equation, as proposed in Refs. [21, 22]).

The Euclidean action for $d=3$ non-Abelian Chern–Simons theory with a $\text{tr } F_{ij}^2$ regulating term reads (compare with the topologically massive Yang–Mills theory [23]):

$$S = S_{\text{CS}} + S_A \quad (2.1)$$

with:

$$S_{\text{CS}} = -\frac{i\kappa}{8\pi} \text{tr} \int_{M_3} d^3x \varepsilon_{ijk} \left(A_i \partial_j A_k + \frac{2}{3} e A_i A_j A_k \right) \quad (2.2)$$

$$S_A = \frac{1}{4A^2} \text{tr} \int_{M_3} d^3x F_{ij}^2. \quad (2.3)$$

Here A_i ($i=1, 2, 3$) takes values in the Lie algebra of the gauge group G , with generators t^a ($\text{tr } t^a t^b = \delta^{ab}$), $\kappa = ke^2$ with $k \in \mathbb{Z}$, and M_3 is the three-dimensional compact manifold. In Eq. (2.3), A is a regulator which will be set to infinity at the end of the computations. This type of regulating Yang–Mills term has already been used [24] to study renormalization of the non-Abelian Chern–Simons theory. It is important to stress that for finite A the model becomes metric-dependent. Consequently it is important to determine whether the resulting effective action does or does not depend on the metric at the quantum level (i.e., whether the quantized theory maintains diffeomorphism invariance, a symmetry of the original Chern–Simons theory). In fact, Chen *et al.* [24] have argued that diffeomorphism anomaly is indeed absent and our results confirm this fact.

We take as the Langevin equation associated with action (2.1)

$$\partial_t A_i(x, t) = -\frac{\delta S}{\delta A_i}(x, t) + \eta_i(x, t), \quad (2.4)$$

where t is the stochastic time ($t \in I \equiv [0, T]$) and η_i is a gaussian noise taking values in the Lie algebra of the group G . A drift term of the form $D_i \Omega$ (with D_i the covariant derivative and Ω an arbitrary function) can be introduced as a way of (stochastic) gauge fixing [25]. Moreover, in the analysis of the connection between $d=3$ Chern–Simons and $d=4$ TYM theories, drift terms naturally appear when one derives Langevin equations by imposing commutativity between BRST transformations and stochastic time evolution [10]. Also, the drift term provides in this context a natural way for introducing an A_0 component for the gauge field in the route towards the effective four-dimensional theory.

In the SQ approach [12], vacuum expectation values (vev) are computed as the limit—for stochastic time going to infinity—of stochastic expectation values:

$$\lim_{T \rightarrow \infty} \langle F[A_\eta] \rangle_\eta = \langle F[A] \rangle. \quad (2.5)$$

Here A_η denotes the solution of Langevin equation giving A as a functional of the

noise; the r.h.s. of Eq. (2.5) is the vev of the functional $F[A]$ for the quantum field theory defined from action (2.1). Concerning stochastic expectation values $\langle \rangle_\eta$, they are computed from the noise correlation functions:

$$\begin{aligned} \langle \eta_i^a \rangle_\eta &= 0 \\ \langle \eta_i^a(x_1, t_1) \eta_j^b(x_2, t_2) \rangle_\eta &= 2\delta^{ab} \delta_{ij} \delta^{(3)}(x_1 - x_2) \delta(t_1 - t_2). \end{aligned} \quad (2.6)$$

Using the explicit form (2.1)–(2.3) for S , Langevin equation in momentum space becomes

$$\dot{A}_i^a = -\frac{1}{\Lambda^2} k^2 P_{ij}(k) A_j^a + \frac{\kappa}{4\pi} \varepsilon_{ijl} k_l A_j^a + Y_i^a(k, t), \quad (2.7)$$

where $\dot{A}_i^a = \partial A_i^a / \partial t$ and $P_{ij}(k)$ and $Y_i^a(k, t)$ are defined as

$$\begin{aligned} P_{ij}(k) &= \delta_{ij} - \frac{k_i k_j}{k^2}, \\ Y_i^a(k, t) &= \eta_i^a(k, t) + \frac{e}{(2\pi)^{3/2}} \int d^3p d^3q \delta(k - p - q) \\ &\quad \times V_{ijk}^{abc}(k, -p, -q) A_j^b(p, t) A_k^c(q, t) \\ &\quad + \frac{e^2}{(2\pi)^3} \int d^3p d^3q d^3r \delta(k - p - q - r) \\ &\quad \times W_{ijkl}^{abcd} A_j^b(p, t) A_k^c(q, t) A_l^d(r, t). \end{aligned} \quad (2.8)$$

In the last equation V_{ijk}^{abc} and W_{ijkl}^{abcd} are, respectively, the three-point and four-point vertex factors (listed in Table I, together with the corresponding diagrams). Note that the three-point vertex factor is just half of the corresponding one in the conventional Feynman rule, while the four-point vertex factor is one-sixth of the corresponding conventional one.

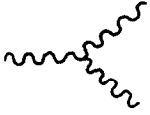
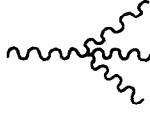
The Langevin equation can be solved perturbatively, and for this purpose it is convenient to write it as an integral equation. Using $A_i^a(k, 0) = 0$ as an initial condition, one readily obtains

$$A_i^a(k, t) = \int_0^\infty dt' G_{ij}^{ab}(k, t - t') Y_j^b(k, t'), \quad (2.10)$$

where $G_{ij}^{ab}(k, t - t')$ is the zeroth-order (causal) Green function associated with Eq. (2.7),

$$\delta^{ab} \left(\eta_{ij} \frac{\partial}{\partial t} + \frac{k^2}{\Lambda^2} P_{ij}(k) - \frac{\kappa}{4\pi} \varepsilon_{ijl} k_l \right) G_{jm}^{bc}(k, t - t') = \delta^{ac} \delta_{im} \delta(t - t'), \quad (2.11)$$

TABLE I
Diagrams for stochastic vertices

DIAGRAM	NOTATION	FORMULA
	$eV_{ijk}^{abc}(k, k_1, k_2)$	$-\frac{i}{2}e f^{abc}[\frac{\kappa}{4\pi}\epsilon_{ijk} + \frac{1}{\Lambda^2}((k - k_1)_k \delta_{ij} + (k_1 - k_2)_i \delta_{jk} + (k_2 - k)_j \delta_{ik})]$
	$e^2 W_{ijkl}^{abcd}$	$-\frac{1}{6}\frac{e^2}{\Lambda^2}[f^{abe}f^{cde}(\delta_{ik}\delta_{jl} - \delta_{il}\delta_{jk}) + f^{ace}f^{bde}(\delta_{ij}\delta_{kl} - \delta_{il}\delta_{jk}) + f^{ade}f^{cbe}(\delta_{ik}\delta_{jl} - \delta_{ij}\delta_{kl})]$

which yields

$$G_{ij}^{ab}(k, t - t') = \theta(t - t') \delta^{ab} \left\{ \left[P_{ij}(k) \cos\left(\frac{\kappa}{4\pi} k(t - t')\right) + \frac{\epsilon_{ijl} k_l}{k} \sin\left(\frac{\kappa}{4\pi} k(t - t')\right) \right] e^{-(1/\Lambda^2) k(t - t')} + \frac{k_i k_j}{k^2} \right\}. \quad (2.12)$$

It is convenient to write Eq. (2.10) in a symbolic way as

$$A = G(\eta + eVAA + e^2WAAA). \quad (2.13)$$

The perturbative expansion of the solution then reads:

$$A = G\eta + eGV(G\eta)(G\eta) + e^2W(G\eta)(G\eta)(G\eta) + e^2GV(GV(G\eta)(G\eta))(G\eta) + e^2GV(G\eta)(GV(G\eta)(G\eta)) + \dots \quad (2.14)$$

which is graphically represented in Fig. 1.

In order to compute an n -point correlation function $\langle A_i^{a_1}(k_1, t) \cdots A_i^{a_n}(k_n, t) \rangle_\eta$, one proceeds as follows: for each $A_i^a(k, t)$ one draws a diagrammatical expansion like that shown in Fig. 1, then one takes the average over the noise by joining all crosses (representing the noise sources), pair-wise, in all possible ways and finally one sums up the resulting "stochastic diagrams," as implied by the noise correlation functions (see Eq. (2.6)).



FIG. 1. Perturbative expansion of $A_i^a(k, t)$.

Note that every diagram is built up from two kinds of propagators:

- (a) the un-crossed propagator given by the stochastic Green function (2.12),

$$G_{ij}^{ab}(k, t-t') = \theta(t-t') \delta^{ab} \left\{ \left[P_{ij}(k) \cos(mk(t-t')) + \frac{\varepsilon_{ijl} k_l}{k} \sin(mk(t-t')) \right] e^{-(1/A^2)k(t-t')} + \frac{k_i k_j}{k^2} \right\}; \quad (2.15)$$

- (b) the crossed propagator which corresponds to the lowest order term in $\langle A_i^a(k, t) A_j^b(-k, t') \rangle_\eta$,

$$\begin{aligned} D_{ij}^{ab}(k; t, t') &= \delta^{ab} 2t_{\min} \frac{k_i k_j}{k^2} + \delta^{ab} \frac{1}{k^2/A^4 + m^2} \\ &\times \left\{ e^{-(k^2/A^2)|t-t'|} \left[\left(\frac{1}{A^2} P_{ij}(k) + \frac{m}{k} \frac{\varepsilon_{ijl} k_l}{k} \right) \cos(mk|t-t'|) \right. \right. \\ &\quad \left. \left. + \left(-\frac{m}{k} P_{ij}(k) + \frac{1}{A^2} \frac{\varepsilon_{ijl} k_l}{k} \right) \sin(mk|t-t'|) \right] \right. \\ &\quad \left. - e^{-(k^2/A^2)(t+t')} \left[\left(\frac{1}{A^2} P_{ij}(k) + \frac{m}{k} \frac{\varepsilon_{ijl} k_l}{k} \right) \cos(mk(t+t')) \right. \right. \\ &\quad \left. \left. + \left(-\frac{m}{k} P_{ij}(k) + \frac{1}{A^2} \frac{\varepsilon_{ijl} k_l}{k} \right) \sin(mk(t+t')) \right] \right\}. \quad (2.16) \end{aligned}$$

In the last equation m stands for $\kappa/4\pi$, usually called the “topological mass” of the model [23], and $t_{\min}$ is the minimum between t and t' . From the above expressions one recognizes a linearly divergent longitudinal part which is common in the SQ of all gauge theories without gauge fixing. This term can be handled by stochastic gauge fixing [25] (see below).

If not for the presence of two kinds of propagators, each stochastic diagram has the form of an ordinary Feynman diagram. Though the rules for these stochastic propagators are different from the Feynman rules, it has been proven for several models that the sum of all stochastic diagrams with the same topology yields exactly the usual result for the corresponding Feynman diagram. In particular, Namiki *et al.* [26] have shown that, up to second order, this is the case for pure Yang–Mills theory. In the same spirit we shall study the second-order corrections to the propagator of the non-Abelian Chern–Simons model (with the $\text{tr } F_{ij}^2$ regulator term) and we shall show that its transverse part reproduces the correct field-theoretical propagator.

As stressed above, correlation functions $\langle A_i^a(k_1, t) A_j^b(k_2, t') \rangle$ contain terms which diverge as t and t' go to infinity. These divergent terms are characteristic of gauge non-invariant quantities and will cancel out if one constructs gauge-invariant

objects like $\langle \text{tr } F_{ij}^2 \rangle$. Consequently we shall only keep those terms which remain finite as t or t' go to infinity. Using (2.14) and averaging over η we then obtain for the correlation function, up to second order,

$$\langle A_i^a(k, t) A_j^b(k', t) \rangle = \delta(k + k') \mathcal{D}_{ij}^{ab}(k, t) \quad (2.17)$$

with

$$\mathcal{D}_{ij}^{ab}(k, t) = (a) + 2(b) + 2(c) + 2(d) + 3(e) + (f), \quad (2.18)$$

where (a), (b), ..., (f) stand for the contributions from diagrams shown in Fig. 2. As an example, diagram (b) represents the following contribution:

$$\begin{aligned} 2(b) = & \frac{2e^2}{(2\pi)^3} \int d^3k_1 d^3k_2 \delta(k - k_1 - k_2) V_{klm}^{cde}(k, -k_1, -k_2) V_{qnp}^{fgh}(-k, k_1, k_2) \\ & \times \int_0^\infty dt' \int_0^\infty dt'' G_{ik}^{ac}(k, t - t') D_{ln}^{dg}(k_1; t', t'') \\ & \times D_{mp}^{eh}(k_2; t', t'') G_{jq}^{bf}(-k, t - t''). \end{aligned} \quad (2.19)$$

Here it should be noted that longitudinal components of external lines can be discarded either because they vanish or because they cancel out when computing gauge invariant or gauge covariant contributions. This is carefully shown in Ref. [26] for the pure Yang-Mills model and the demonstration trivially applies to the present model. Concerning longitudinal components appearing in internal lines,

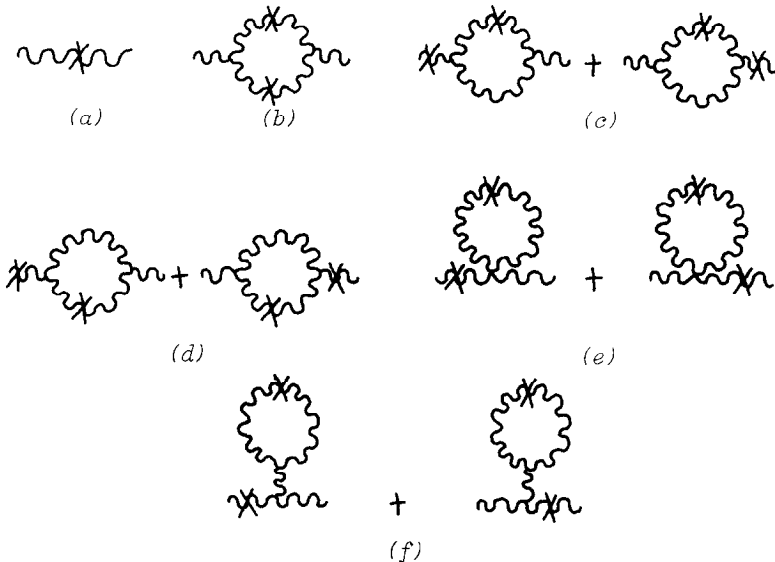


FIG. 2. Graphical representation of $\mathcal{D}_{ij}^{ab}(k, t)$.

it has been conjectured by Parisi and Wu [12] that they arrange themselves to give just those contributions which are conventionally associated with Faddeev–Popov ghost effects. Although a general proof for this conjecture is lacking, it has been explicitly confirmed in several examples [27]. In particular, Namiki *et al.* [26] have proved that, up to second order, in the case of pure Yang–Mills theory internal lines arrange to reproduce the ghost contributions. Since the longitudinal part of the propagators of this last theory coincide with those arising in the CS model (Eqs. (2.16) and (2.17)), we conclude that, also in the present case, ghost effects are accounted for by longitudinal parts of internal lines and hence we shall only consider in diagrams (a), (b), ..., (f) transverse internal lines.

Computation of contributions from different diagrams is standard but tedious. For the sake of brevity we shall not present each one separately. Just as an example, the contribution from diagram (b) takes the form:

$$2(b) = \frac{e^2}{(2\pi)^3} \int d^3k_1 d^3k_2 \delta(k - k_1 - k_2) V_{klm}^{ade}(k, -k_1, -k_2) V_{qnp}^{bde}(-k, k_1, k_2) \\ \times I_{iklnmpqj}(k, k_1, k_2, m, \Lambda), \quad (2.20)$$

where

$$I_{iklnmpqj}(k, k_1, k_2, m, \Lambda) = P_{ik}(k) P_{ln}(k_1) P_{mp}(k_2) P_{jq}(k) F^{(1)} \\ + P_{ik}(k) P_{ln}(k_1) P_{mp}(k_2) \frac{\varepsilon_{jqv} k_v}{k} F^{(2)} \\ + P_{ik}(k) P_{ln}(k_1) \frac{\varepsilon_{mpu} k_{2u}}{k_2} P_{jq}(k) F^{(3)} \\ + P_{ik}(k) P_{ln}(k_1) \frac{\varepsilon_{mpu} k_{2u}}{k_2} \frac{\varepsilon_{jqv} k_v}{k} F^{(4)} \\ + P_{ik}(k) \frac{\varepsilon_{lns} k_{1s}}{k_1} P_{mp}(k_2) P_{jq}(k) F^{(5)} \\ + P_{ik}(k) \frac{\varepsilon_{lns} k_{1s}}{k_1} P_{mp}(k_2) \frac{\varepsilon_{jqv} k_v}{k} F^{(6)} \\ + P_{ik}(k) \frac{\varepsilon_{lns} k_{1s}}{k_1} \frac{\varepsilon_{mpu} k_{2u}}{k_2} P_{jq}(k) F^{(7)} \\ + P_{ik}(k) \frac{\varepsilon_{lns} k_{1s}}{k_1} \frac{\varepsilon_{mpu} k_{2u}}{k_2} \frac{\varepsilon_{jqv} k_v}{k} F^{(8)} \\ + \frac{\varepsilon_{ikr} k_r}{k} P_{ln}(k_1) P_{mp}(k_2) P_{jq}(k) F^{(9)}$$

$$\begin{aligned}
& + \frac{\varepsilon_{ikr} k_r}{k} P_{ln}(k_1) P_{mp}(k_2) \frac{\varepsilon_{jqv} k_v}{k} F^{(10)} \\
& + \frac{\varepsilon_{ikr} k_r}{k} P_{ln}(k_1) \frac{\varepsilon_{mpu} k_{2u}}{k_2} P_{jq}(k) F^{(11)} \\
& + \frac{\varepsilon_{ikr} k_r}{k} P_{ln}(k_1) \frac{\varepsilon_{mpu} k_{2u}}{k_2} \frac{\varepsilon_{jqv} k_v}{k} F^{(12)} \\
& + \frac{\varepsilon_{ikr} k_r}{k} \frac{\varepsilon_{lns} k_{1s}}{k_1} P_{mp}(k_2) P_{jq}(k) F^{(13)} \\
& + \frac{\varepsilon_{ikr} k_r}{k} \frac{\varepsilon_{lns} k_{1s}}{k_1} P_{mp}(k_2) \frac{\varepsilon_{jqv} k_v}{k} F^{(14)} \\
& + \frac{\varepsilon_{ikr} k_r}{k} \frac{\varepsilon_{lns} k_{1s}}{k_1} \frac{\varepsilon_{mpu} k_{2u}}{k_2} P_{jq}(k) F^{(15)} \\
& + \frac{\varepsilon_{ikr} k_r}{k} \frac{\varepsilon_{lns} k_{1s}}{k_1} \frac{\varepsilon_{mpu} k_{2u}}{k_2} \frac{\varepsilon_{jqv} k_v}{k} F^{(16)}. \quad (2.21)
\end{aligned}$$

Here $F^{(i)} = F^{(i)}(k, k_1, k_2, m, A)$ are ratios of polynomials in momenta which arise after integrating the fictitious time in the vertices and taking the limit $t \rightarrow \infty$. Contributions from the rest of the diagrams having the same topology as (b) have analogous forms. Computations were carried out using REDUCE and the answer is

$$\begin{aligned}
2(b) + 2(c) + 2(d) &= 2 \frac{e^2}{(2\pi)^3} \int d^3 k_1 d^3 k_2 \delta(k - k_1 - k_2) \\
&\times V_{klm}^{ade}(k, -k_1, -k_2) V_{bde}^{fgh}(-k, k_1, k_2) \\
&\times \frac{((1/A^2) P_{ik}(k) + (m/k^2) \varepsilon_{ikr} k_r)}{k^2/A^4 + m^2} \\
&\times \frac{((1/A^2) P_{ln}(k_1) + (m/k_1^2) \varepsilon_{lns} k_{1s})}{k_1^2/A^4 + m^2} \\
&\times \frac{((1/A^2) P_{mp}(k_2) + (m/k_2^2) \varepsilon_{mpu} k_{2u})}{k_2^2/A^4 + m^2} \\
&\times \frac{((1/A^2) P_{jq}(k) + (m/k^2) \varepsilon_{jqv} k_v)}{k^2/A^4 + m^2}. \quad (2.22)
\end{aligned}$$

An analogous calculation for diagrams (e) gives

$$\begin{aligned}
 3(e) = & 3 \frac{e^2}{(2\pi)^3} \int d^3 k_1 W_{klmn}^{ad\bar{d}b} \frac{((1/\Lambda^2) P_{ik}(k) + (m/k^2) \varepsilon_{ikr} k_r)}{k^2/\Lambda^4 + m^2} \\
 & \times \frac{((1/\Lambda^2) P_{lm}(k_1) + (m/k_1^2) \varepsilon_{lmu} k_u)}{k_1^2/\Lambda^4 + m^2} \\
 & \times \frac{((1/\Lambda^2) P_{jn}(k) + (m/k^2) \varepsilon_{jns} k_s)}{k^2/\Lambda^4 + m^2}. \quad (2.23)
 \end{aligned}$$

Taking into account the relations between stochastic and conventional Feynman vertices, it is now evident that Eqs. (2.22) and (2.23) coincide, respectively, with the gluon loop and the tadpole terms in the conventional Landau gauge field theory (see, for instance, [28]). Finally, diagrams (f) vanish due to antisymmetry of the three-point vertex. Hence we arrive at

$$(a) + 2(b) + 2(c) + 2(d) = \text{wavy line} + \text{gluon loop} \quad (2.24)$$

$$3(e) + (f) = \text{gluon tadpole} + \text{gluon tadpole} \quad (2.25)$$

Here wavy lines represent the standard (bare) propagator for topologically massive gauge theory, i.e., $d=3$ Yang–Mills theory plus a CS term (compare e.g., Ref. [28]):

$$\text{wavy line} = \frac{(\frac{1}{\Lambda^2} P_{ij}(k) + \frac{m}{k^2} \varepsilon_{ijr} k_r)}{\frac{k^2}{\Lambda^4} + m^2} \quad (2.26)$$

Thus, the r.h.s. in (2.24) and (2.25) reproduce the standard second-order result in perturbation theory. Concerning the resulting three-point and four-point vertices, they coincide with the standard vertices for topologically massive gauge theory and they can be read off from those presented in Table I (provided we disregard the factors $\frac{1}{2}$ and $\frac{1}{6}$, respectively, appearing in the stochastic scheme).

We can therefore conclude that, up to second order in perturbation, the stochastic quantization scheme for the non-Abelian CS model, endowed with a regulator term of the form (2.3), leads to the correct gauge field propagator for topologically massive Yang–Mills theory. That is, the introduction of the $(1/\Lambda^2) \text{tr} F_{ij}^2$ term has led to a stochastic process which converges towards equi-

librium. As we have already stated, diagrams with ghost lines should be obtained by including the longitudinal parts of G_{ij}^{ab} and D_{ij}^{ab} in the internal lines of each diagram.

If one now takes the limit of $\Lambda^2 \rightarrow \infty$ to switch off the regulator, so as to recover the pure CS gauge theory, one can easily verify that (2.24)–(2.26) agree with the standard (bare) expressions (compare, e.g., Ref. [29]). We have then succeeded in obtaining a convergent stochastic process for the non-Abelian CS theory.

We are now in position to re-analyze the connection between the (regulated) stochastic partition function for CS theory and the BRST partition function for TYM theory. The connection between $d=3$ non-Abelian CS and $d=4$ TYM theories has been established by Yu [9] and Baulieu [10] using SQ. (Alternatively, Horowitz [7] has discussed the same connection using canonical quantization for systems with constraints.)

Concerning the SQ derivation, it consists in proving the equivalence between the stochastic partition function for CS theory in the limit $T \rightarrow \infty$ and the BRST partition function for TYM theory (stochastic time t provides the extra coordinate necessary to pass from M_3 to a $d=4$ manifold M_4).

An important point that was not taken into account in this derivation is that related to the non-convergence of the underlying stochastic process. Now, we have proven above that the introduction of a regulator in the Langevin equation yields the correct equilibrium limit for the stochastic process and gives the standard propagators for the non-Abelian CS theory. With this in mind, one can re-examine the question of whether the connection between CS and TYM theories is still valid when the theory is regulated. This can be very easily done following the steps in Refs. [9–11], but including the regulator term. One ends with a stochastic partition function for the Chern–Simons theory in the form

$$Z_{\text{Stoch}}^{\text{CS}}[A] = \int D\Phi e^{-S_{\text{eff}}}, \quad (2.27)$$

where

$$S_{\text{eff}} = \{Q, V_A\} \quad (2.28)$$

and

$$\begin{aligned} V_A = \int_{M_4} d^4x \, \text{tr} \left\{ \frac{1}{4} \chi_{\alpha\beta} \left(F_{\alpha\beta}^+ - \frac{1}{2} \eta_{\alpha\beta} \right) - \chi_{0i} \frac{4\pi}{\kappa \Lambda^2} D_j F_{ji} \right. \\ \left. + b \left(\partial_\mu A_\mu - \frac{1}{2} \eta \right) - \lambda (D_\mu \psi_\mu + D_\mu D_\mu c) \right\}. \end{aligned} \quad (2.29)$$

In Eq. (2.27) we have called Φ the whole field content,

$$\Phi \equiv (A_\mu, b, \chi_{\alpha\beta}, c, \lambda, \psi_\mu, \eta, \eta_{\alpha\beta}, \phi, \rho) \quad (2.30)$$

which includes the original gauge field as well as Lagrange multipliers $\eta_{\alpha\beta}$ and η and ghost fields $b, \chi_{\alpha\beta}, c, \lambda, \psi_\mu, \phi, \rho$ that are necessary to write $Z_{\text{Stoch}}^{\text{CS}}[A]$ in the form (2.27) (see Ref. [11] for details). In Eq. (2.28), $\{Q, \}$ represents a BRST transformation defined as

$$\{Q, \Phi\} = (\psi_\mu, \eta, \eta_{\alpha\beta}, \phi, \rho, 0, 0, 0, 0, 0). \quad (2.31)$$

Inspection of Eqs. (2.27)–(2.29) shows that $Z_{\text{Stoch}}^{\text{CS}}[A]$ coincides with the BRST partition function for TYM theory, except for the A -dependent terms. We write this connection in the form:

$$Z_{\text{Stoch}}^{\text{CS}}[A] = Z_{\text{BRST}}^{\text{TYM}}[A]. \quad (2.32)$$

Since the A -dependent terms in S_{eff} appear through Q -exact terms one can easily infer that $Z_{\text{BRST}}^{\text{TYM}}[A]$ is, in fact, A -independent, so it coincides with the usual partition function for TYM theory. This does not mean that from the stochastic point of view the regulator is not necessary. As already explained, in order to compute CS propagators, etc., one has to maintain the regulator until the end of the computations. This means that one can use a regulator A to render all quantities appearing in the derivation of (2.27) well defined; at the end of the computations, being the resulting partition function A -independent, one can make $A \rightarrow \infty$, thus confirming the connection between CS and TYM models.

Several other topological features can be easily established for the partition function (2.32). In particular, $Z_{\text{BRST}}^{\text{TYM}}$ defines a topological model for *any* value of A ; it does not depend on the coupling constant e , so that one can compute it in the weak-coupling (semi-classical) limit; finally the A -independence of $Z_{\text{BRST}}^{\text{TYM}}$ extends to any (topological invariant) observable. It is also worthwhile to note that the partition function will yield the first Donaldson invariant, provided one takes the $A \rightarrow \infty$ limit.

At this point it is interesting to make contact with the results of Chen *et al.* [24] on topological features in perturbative CS theory. Using both dimensional and $\text{tr } F_{ij}^2$ regularizations, these authors showed the vanishing of the β function for CS theory up to three loops and also suggested the absence of the diffeomorphism anomaly (which could be present due to the metric dependence of the $\text{tr } F_{ij}^2$ regulator). Indeed, by studying the propagator for the regulated CS model they showed that its metric-dependent transverse part (responsible for the would-be diffeomorphism anomaly) vanishes because of the unusual properties of the wave-function renormalization. Our results, summarized by Eq. (2.32), confirm this suggestion.

We conclude this discussion by analysing an alternative to the use of a $(1/A^2) \text{tr } F_{ij}^2$ regulator, namely, the introduction of a non-trivial kernel K_{ij} in the Langevin equation in order to have a stochastic process which converges to equilibrium. In fact, this alternative approach was followed in Refs. [21, 22] in the

study of the Abelian and non-Abelian Chern–Simons models. In this last case, if one chooses K_{ij} in the form

$$K_{ij} = -\varepsilon_{ijk} \partial_k, \quad (2.33)$$

then the generalized Langevin equation,

$$\dot{A}_i^a = K_{ij} \frac{\delta S}{\delta A_j^a} + \eta_i^a, \quad (2.34)$$

yields the correct equilibrium limit *to all orders in perturbation theory* so that conventional propagators are nicely obtained [21]. Now, it is easy to prove, using the approach presented in Ref. [7], the CS–TYM connection in the presence of a kernel. The interesting fact is that, in this approach, convergence can be proven to all orders in perturbation theory [21].

3. CONCLUSIONS

The original studies [9–11] of the CS–TYM connection using SQ were somehow formal in the sense that the non-convergence of the underlying stochastic process was disregarded. Later on, the problem of convergence in SQ of CS theory was solved [21, 22] by introducing appropriate kernels or regulators so that a correct equilibrium limit can be attained. After these results, it remained to determine whether the connection between CS and TYM theories was still valid in the presence of regulators.

In the course of such a study we have found it convenient to analyse, in the same vein as Namiki *et al.* [26] did for the SQ of Yang–Mills theory, how convergence for the non-Abelian CS model is attained when the regulator is introduced and the precise way in which standard propagators are obtained using SQ. We have proven that the addition of a regulator $(1/\Lambda^2) \text{tr} F_{ij}^2$ to the CS action renders the stochastic process associated with the CS Langevin equation convergent. The proof we have presented is valid up to second order in perturbation theory and states that the transverse part of the stochastic diagrams reproduces the standard gluonic CS propagator (see Eq. (2.24)). It is interesting to note that the way in which stochastic diagrams for CS regulated theory sum up to the standard propagator is basically identical to the way it works in the case of pure YM theory.

We have also considered the alternative regularization scheme which consists in the introduction of a non-trivial kernel. Indeed, an appropriate choice of the kernel [21, 22] (see Eq. (2.33)) yields the correct equilibrium limit *to all orders in perturbation theory*.

We have explained how, in both regularization approaches, the equivalence between the stochastic partition function for CS theory and the BRST partition function for TYM theory can be proved in the presence of a regulator.

REFERENCES

1. E. WITTEN, *Commun. Math. Phys.* **117** (1988), 353.
2. E. WITTEN, *Commun. Math. Phys.* **118** (1989), 411.
3. M. DUNE, R. JACKIW, AND M. TRUGENBERGER, *Ann. Phys. (N.Y.)* **194** (1989), 197.
4. M. BLAU, D. BIRMINGHAM, M. RAKOWSKI, AND G. THOMPSON, *Phys. Rep.* **209** (1991), 129.
5. J. LABASTIDA AND PERNICI, *Phys. Lett. B* **212** (1988), 91.
6. D. BIRMINGHAM, M. RAKOWSKI, AND G. THOMPSON, *Phys. Lett. B* **212** (1988), 187; **214** (1988), 381.
7. G. HOROWITZ, *Commun. Math. Phys.* **125** (1989), 417.
8. L. BAULIEU AND B. GROSSMANN, *Phys. Lett. B* **212** (1988), 351.
9. Y. YU, *Phys. Rev. D* **40** (1989), 1301.
10. L. BAULIEU, *Phys. Lett. B* **232** (1989), 479.
11. L. CUGLIANDOLO, G. LOZANO, AND F. A. SCHAPOSNIK, *Phys. Lett. B* **253** (1991), 90.
12. G. PARISI AND Y. S. WU, *Sci. Sinica* **24** (1981), 483.
13. G. PARISI AND N. SOURLAS, *Nucl. Phys. B* **206** (1982), 321.
14. S. CECOTTI AND L. GIRARDELLO, *Ann. Phys. (N.Y.)* **145** (1983), 81.
15. K. FUJIKAWA, *Phys. Rev. Lett.* **42** (1972), 1195; **44** (1980), 1733; *Phys. Rev. D* **38** (1980), 2848.
16. L. CUGLIANDOLO, G. LOZANO, AND F. A. SCHAPOSNIK, *Phys. Lett. B* **244** (1990), 249.
17. R. K. KAUL AND R. RAJARAMAN, *Phys. Lett. B* **249** (1990), 433.
18. G. PARISI, *Phys. Lett. B* **131** (1983), 393.
19. J. AMBJORN AND S. K. YANG, *Phys. Lett. B* **165** (1985), 140.
20. A. POLYKRONAKOS AND R. TZANI, *Phys. Lett. B* **259** (1991), 291.
21. Y. S. WU AND ZHU, University of Utah preprint, 1991.
22. F. FERRARI AND H. HÜFFEL, *Phys. Lett. B* **261** (1991), 47.
23. R. JACKIW AND S. TEMPLETON, *Phys. Rev. D* **23** (1981), 2291; S. DESER, R. JACKIW, AND S. TEMPLETON, *Phys. Rev. Lett.* **49** (1982), 975; *Ann. Phys. (N.Y.)* **140** (1982), 372.
24. W. CHEN, G. W. SEMENOFF, AND Y. S. WU, *Mod. Phys. Lett. A* **23** (1990), 1833.
25. D. ZWANZIGER, *Nucl. Phys. B* **192** (1981), 259.
26. M. NAMIKI, I. OHBA, K. OKANO, AND Y. YAMANAKA, *Prog. Theor. Phys.* **69** (1983), 1580.
27. P. H. DAMGAARD AND H. HÜFFEL, *Phys. Rep.* **152** (1987), 227.
28. R. PISARSKI AND S. RAO, *Phys. Rev. D* **22** (1985), 2081.
29. ALVAREZ-GAUME, LABASTIDA, AND RAMALLO, *Nucl. Phys. B* **334** (1990), 103.